

Ordered Fractional Integration (OFI)

A Complete Treatment:

From $(-1)!$ and the Gamma Function
to the Standard Model Lagrangian and Atomic Orbitals

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2026-03-28

Contents

1 Introduction

Classical calculus gives derivatives of integer order: $f', f'', f''', \dots$ and integrals $\int f, \iint f, \dots$. A natural question is: *what is a derivative of order $\frac{1}{2}$? Or of order π ? Of order -1 ?*

This paper builds the answer from the ground up, requiring only standard calculus as a prerequisite. Along the way we encounter:

- Why $(-1)! = \infty$ is not an error but a mechanism.
- Why the “obvious” fractional integral (Riemann–Liouville) has a fatal non-commutativity flaw — the *ordering problem*.
- How the **Ordered Fractional Integral (OFI)**, defined via the symmetric Riesz kernel, resolves every flaw at once.
- Why the regularized divergent sum $1 + 2 + 3 + \dots$ carries *two* pieces of information, written $(-1, \frac{1}{12})$, not just $-\frac{1}{12}$.
- How the 30+ extended trigonometric functions collapse into a single OFI orbit under fractional differentiation.
- How OFI applied to the Standard Model Lagrangian leads directly to the Yang–Mills mass-gap problem.
- How fractional calculus models atomic hybridization, ionic vs. covalent bonding, and nucleon–anti-nucleon structure via gluons.

A note on prior errors. An earlier version of this framework incorrectly identified the OFI defect with a boxcar (moving-average) filter. That is *not* fractional calculus — it is statistics. This paper uses only the genuine Riesz fractional integral throughout. The word “fractional” is load-bearing everywhere it appears.

2 The Gamma Function: Extending Factorials to Every Number

2.1 The factorial problem

For positive integers, $n! = n(n-1)(n-2)\dots 1$. Can we define $z!$ for non-integer or negative z ?

Definition 2.1 (Gamma function). For $\text{Re}(z) > 0$:

$$\Gamma(z) := \int_0^\infty t^{z-1} e^{-t} dt.$$

The key relation is $\Gamma(z+1) = z\Gamma(z)$, proved by integration by parts. Since $\Gamma(1) = 1$, this gives $\Gamma(n) = (n-1)!$ for all positive integers n . So Γ is the factorial, shifted by one:

$$n! = \Gamma(n+1).$$

2.2 Analytic continuation and poles

The integral definition only converges for $\text{Re}(z) > 0$. But the functional equation $\Gamma(z) = \Gamma(z+1)/z$ extends $\Gamma(z)$ to all of $\mathbb{C}$ *except* $z = 0, -1, -2, -3, \dots$, where it has simple poles.

z	$\Gamma(z)$	Meaning
5	24	$4! = 24$
3	2	$2! = 2$
2	1	$1! = 1$
1	1	$0! = 1$ (convention confirmed)
$\frac{1}{2}$	$\sqrt{\pi}$	$(-\frac{1}{2})! = \sqrt{\pi}$
0^+	$+\infty$	pole: $(-1)! = \infty$
$-\frac{1}{2}$	$-2\sqrt{\pi}$	finite, negative
-1^+	$-\infty$	pole: $(-2)! = \infty$
$-\frac{3}{2}$	$\frac{4}{3}\sqrt{\pi}$	finite, positive

2.3 Why $(-1)! = \infty$ is correct, not broken

Near $z = 0$, the Laurent expansion is:

$$\Gamma(z) = \frac{1}{z} - \gamma + O(z),$$

where $\gamma \approx 0.5772$ is the Euler–Mascheroni constant. The residue at $z = 0$ is 1, confirming a simple pole.

This is the *mechanism* behind a deep fact: as $\alpha \rightarrow 0^+$, the Riesz kernel $c_{n,\alpha}|x - y|^{\alpha-n} \rightarrow \delta(x - y)$ (the Dirac delta), so the fractional integral $I^\alpha \rightarrow \text{Id}$ (identity). The $1/z$ pole in Γ and the non-integrable singularity $|x - y|^{-n}$ *cancel exactly*, forcing the identity at $\alpha = 0$. The infinity is the mechanism, not the error.

2.4 The $(-1, \frac{1}{12})$ representation of regularized divergences

The sum $S = 1 + 2 + 3 + \dots$ diverges. Classical regularization via the Riemann zeta function gives $\zeta(-1) = -\frac{1}{12}$. But *reporting only $-\frac{1}{12}$ discards information*.

The zeta function has a simple pole at $s = 1$ with residue 1:

$$\zeta(s) = \frac{1}{s-1} + \gamma + O(s-1).$$

The full information about S is the pair

$$\boxed{(\text{pole order, regularized value}) = (-1, \frac{1}{12})}.$$

Here:

- -1 encodes the *location of the divergence* (the pole of ζ at $s = 1$, shifted by 2 to reach $s = -1$, the evaluation point).
- $\frac{1}{12}$ is the *regularized finite part*: $|\zeta(-1)| = \frac{1}{12}$.

In the OFI framework, a regularized quantity is always a *pair* (analytic structure, finite value) — just as a complex number carries both magnitude and phase. Reporting only $-\frac{1}{12}$ is like reporting only the magnitude of a complex number.

2.5 The discrete–continuous bridge: why $\pm\frac{1}{2}$ is exact

The $\frac{1}{2}$ that appears in $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, in the regularization pair $(-1, \frac{1}{12})$, and throughout fractional calculus is not a coincidence. It is the same $\frac{1}{2}$ that bridges discrete sums and continuous integrals.

The fundamental gap. For any continuous function f , the continuous integral and the discrete sum are *not equal* over the same nominal bounds:

$$\int_a^b f(x) dx \neq \sum_{k=a}^b f(k).$$

The integral is systematically *short*: it misses a half-unit of area at each endpoint because it evaluates f everywhere, while the sum evaluates f only at integers.

The fix: three equivalent forms. To make them exactly equal, choose one of:

Form	Rule	How to read it
Bound shift	$\int_{a-1/2}^{b+1/2} f(x) dx = \sum_{k=a}^b f(k)$	widen both bounds by half a unit
Lower shift	$\int_a^{b+1} f(x - \frac{1}{2}) dx = \sum_{k=a}^b f(k)$	shift integrand left by $\frac{1}{2}$
Upper shift	$\int_{a-1}^b f(x + \frac{1}{2}) dx = \sum_{k=a}^b f(k)$	shift integrand right by $\frac{1}{2}$

All three are equivalent (one substitution away from each other). The rule is: *a discrete sum over integers equals a continuous integral over the half-integer midpoints of the unit cells surrounding each integer.*

Proof for $f(x) = x$ (triangle numbers). The triangle number $T(n) = 1 + 2 + \cdots + n = \frac{n(n+1)}{2}$. The naive integral falls short:

$$\int_1^n x dx = \frac{n^2 - 1}{2} \neq T(n).$$

Apply the bound shift:

$$\int_{1-1/2}^{n+1/2} x dx = \frac{(n + \frac{1}{2})^2 - (\frac{1}{2})^2}{2} = \frac{n^2 + n + \frac{1}{4} - \frac{1}{4}}{2} = \frac{n(n+1)}{2} = T(n). \quad \checkmark$$

Or equivalently, shift the integrand by $+\frac{1}{2}$:

$$\int_0^n (x + \frac{1}{2}) dx = \frac{n^2}{2} + \frac{n}{2} = T(n). \quad \checkmark$$

Both give the exact answer.

Why this is recursive. The derivative of $T(n) = \frac{n(n+1)}{2}$ is $T'(n) = n + \frac{1}{2}$ — a half-integer, not an integer. So the integrand that recovers the discrete sum is *the derivative of the sum itself evaluated at the midpoint*. The $\frac{1}{2}$ is *recycled*: it appears in the derivative of T , and it is also the bound correction needed to make $\int = \sum$. This self-referential $\frac{1}{2}$ is why triangle numbers are “derivative-recursive”: $T'(n) = n + \frac{1}{2}$ evaluated at n gives the next term of the sum, plus a half that carries to the integral correction.

More precisely: integrating the derivative $T'(x) = x + \frac{1}{2}$ from 0 to n recovers $T(n)$ exactly, while integrating the “naïve” integrand x from 0 to n gives $n^2/2$, which is short by exactly $n/2$ — the sum of n copies of $\frac{1}{2}$, one per integer step.

The general rule for bounds a to b . For any $a \leq b$ integers and $f(x) = x$:

$$\int_{a-1/2}^{b+1/2} x \, dx = \frac{b(b+1)}{2} - \frac{a(a-1)}{2} = T(b) - T(a-1) = \sum_{k=a}^b k.$$

The pattern: the lower bound loses $\frac{1}{2}$ (min $-\frac{1}{2}$), the upper bound gains $\frac{1}{2}$ (max $+\frac{1}{2}$).

Connection to OFI. This $\pm\frac{1}{2}$ shift is the *same centering correction* that the Riesz kernel applies. The Riesz kernel $|x-y|^{\alpha-n}$ is symmetric about the midpoint $\frac{x+y}{2}$ — it automatically evaluates at the center of every pair of points, not at either endpoint. The OFI with $\alpha = \frac{1}{2}$ corresponds exactly to the continuous analog of the discrete $\pm\frac{1}{2}$ bound shift: both measure at the *midpoint of a unit cell*, not the boundary.

This is why the natural fractional order for Navier–Stokes is $\alpha = \frac{1}{2}$ (Section 5): it is the order at which the continuous integral aligns with the discrete counting of vorticity quanta.

3 Fractional Calculus: Derivatives Between 0 and 1

3.1 The key idea

A standard derivative shifts a function: $D^1[\sin x] = \cos x$. A second derivative shifts again: $D^2[\sin x] = -\sin x$. Four derivatives return you to the start: $D^4[\sin x] = \sin x$.

Fractional differentiation interpolates:

$$D^\alpha[\sin x] = \sin\left(x + \frac{\alpha\pi}{2}\right), \quad \alpha \in \mathbb{R}.$$

α	$D^\alpha[\sin x]$	Description
0	$\sin x$	Identity
$\frac{1}{4}$	$\sin(x + \pi/8)$	“Quarter derivative”
$\frac{1}{2}$	$\sin(x + \pi/4)$	“Half derivative”
1	$\cos x$	Standard derivative
2	$-\sin x$	Second derivative
4	$\sin x$	Returns to start
−1	$-\cos x$	Integration

Fractional differentiation *rotates* functions in phase space. Every derivative of sin is still a trig function — just phase-shifted. This is not an approximation; it is exact.

3.2 The Riemann–Liouville fractional integral

The first rigorous definition: for $\alpha > 0$,

$$(I_{\text{RL}}^\alpha f)(x) := \frac{1}{\Gamma(\alpha)} \int_0^x (x-t)^{\alpha-1} f(t) dt.$$

Check: as $\alpha \rightarrow 1$, this becomes $\int_0^x f(t) dt$ (ordinary integral).

Check: $I_{\text{RL}}^1 I_{\text{RL}}^1 f = I_{\text{RL}}^2 f$ (semigroup in integers).

The prefactor $1/\Gamma(\alpha)$ is essential. At $\alpha = 0$: $1/\Gamma(0) = 0$ (since $\Gamma(0) = \infty$), so $I_{\text{RL}}^0 f = 0$ — WRONG. The identity should give back f . This is the $(-1)! = \infty$ issue forcing $1/(-1)! = 0$, which kills the integral.

The Riesz approach (Section 5) resolves this.

4 The Ordering Problem in Fractional Calculus

4.1 What is the ordering problem?

In standard calculus, differentiation and integration are inverse operations:

$$\frac{d}{dx} \int_0^x f(t) dt = f(x). \quad (\text{FTC})$$

In Riemann–Liouville fractional calculus:

$$D_{\text{RL}}^\alpha I_{\text{RL}}^\alpha f = f \quad \checkmark$$

but

$$I_{\text{RL}}^\alpha D_{\text{RL}}^\alpha f = f - \frac{f^{(\alpha-1)}(0^+)}{\Gamma(\alpha)} x^{\alpha-1} \neq f \quad \text{in general.}$$

The difference is a boundary correction term involving the value of f at the left endpoint $x = 0$. This means:

$$D_{\text{RL}}^\alpha I_{\text{RL}}^\alpha \neq I_{\text{RL}}^\alpha D_{\text{RL}}^\alpha.$$

The operators do not commute. The order in which you write D^α and I^α matters. This is the **ordering problem**.

4.2 Why it breaks physics

In quantum mechanics and field theory, operators represent physical observables. Non-commutativity of D^α and I^α means:

- “Differentiate then integrate” $\neq$ “integrate then differentiate.”
- The fractional Leibniz rule picks up boundary ghosts.
- The semigroup property fails: $I_{\text{RL}}^\alpha I_{\text{RL}}^\beta \neq I_{\text{RL}}^{\alpha+\beta}$ in general on bounded domains.

The Caputo derivative (a variant) fixes some issues but not all. Neither Riemann–Liouville nor Caputo gives a clean semigroup law on $\mathbb{R}^n$.

5 OFI: The Riesz Ordered Fractional Integral

5.1 The Riesz potential — genuine fractional calculus

Definition 5.1 (Riesz potential / OFI). For $0 < \alpha < n$, $f : \mathbb{R}^n \rightarrow \mathbb{R}$:

$$(I^\alpha f)(x) := c_{n,\alpha} \int_{\mathbb{R}^n} |x - y|^{\alpha-n} f(y) dy,$$

where

$$c_{n,\alpha} = \frac{\Gamma(\frac{n-\alpha}{2})}{2^\alpha \pi^{n/2} \Gamma(\frac{\alpha}{2})}.$$

The kernel $|x - y|^{\alpha-n}$ is:

- **Power-law**: not a box, not a Gaussian, not a moving average.
- **Symmetric**: depends only on the distance $|x - y|$, not on direction.
- **Non-compact**: it reaches everywhere in $\mathbb{R}^n$, with weight decaying as a power law.

In Fourier space, $\widehat{I^\alpha f}(\xi) = |\xi|^{-\alpha} \hat{f}(\xi)$. The Fourier multiplier is $|\xi|^{-\alpha}$ — a genuine fractional power.

5.2 Why “Ordered”: the symmetric kernel resolves the ordering problem

The Riesz kernel $|x - y|^{\alpha-n}$ is symmetric: $|x - y| = |y - x|$. It has no preferred direction and no boundary. Therefore:

$$(-\Delta)^{\alpha/2} I^\alpha = \text{Id} \quad \text{and} \quad I^\alpha (-\Delta)^{\alpha/2} = \text{Id} \quad (\text{up to constants, on } \mathbb{R}^n).$$

The operators commute. The ordering problem disappears.

This is why the integral is called **Ordered**: it solves the ordering problem that afflicts Riemann–Liouville and Caputo.

5.3 The semigroup law

Theorem 5.2 (OFI semigroup). $I^\alpha I^\beta = I^{\alpha+\beta}$ for $0 < \alpha, \beta, \alpha + \beta < n$.

Proof. In Fourier space: $\widehat{I^\alpha I^\beta f}(\xi) = |\xi|^{-\alpha} \cdot |\xi|^{-\beta} \cdot \hat{f}(\xi) = |\xi|^{-(\alpha+\beta)} \hat{f}(\xi) = \widehat{I^{\alpha+\beta} f}(\xi)$. □

Compare the boxcar filter $K_a(x) = \frac{1}{a} \mathbf{1}_{[-a/2, a/2]}(x)$, which has Fourier multiplier $\text{sinc}(a\xi/2)$. Then:

$$K_a \circ K_a \text{ has multiplier } \text{sinc}^2(a\xi/2) \neq \text{sinc}(a\xi/2).$$

The boxcar *fails* the semigroup law. It gives a tent (triangular) kernel, not a boxcar. This failure is the mathematical proof that boxcar averaging is NOT fractional calculus.

5.4 The centering principle

The Riesz kernel is symmetric about the midpoint $\frac{x+y}{2}$. This centering appears throughout mathematics:

Context	Formula	The centering
Triangle numbers	$T(n) = \int_0^n (x + \frac{1}{2}) dx$	$+\frac{1}{2}$ shifts to midpoint
Collatz map	$3n + 1 = 3(n + \frac{1}{3})$	$+\frac{1}{3}$ is the ordered rate
Central difference	$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}$	symmetric about x
Riesz kernel	$ x - y ^{\alpha-n}$	symmetric about $\frac{x+y}{2}$
$\sin \theta$ at small θ	$\sin \theta \approx \theta$	centered at origin

In all cases: evaluate at the **midpoint**, not the endpoint. This eliminates boundary bias and ordering ambiguity simultaneously.

6 Practice Examples

Example 6.1 (Half-derivative of x^n). The Riemann–Liouville half-derivative of $f(x) = x^\beta$ for $\beta > -1$:

$$D_{\text{RL}}^{1/2} x^\beta = \frac{\Gamma(\beta + 1)}{\Gamma(\beta + \frac{1}{2})} x^{\beta-1/2}.$$

Particular cases:

$$\begin{aligned} D^{1/2}[1] &= \frac{\Gamma(1)}{\Gamma(\frac{1}{2})} x^{-1/2} = \frac{1}{\sqrt{\pi}} x^{-1/2}, \\ D^{1/2}[x] &= \frac{\Gamma(2)}{\Gamma(\frac{3}{2})} x^{1/2} = \frac{1}{\Gamma(\frac{3}{2})} \sqrt{x} = \frac{2}{\sqrt{\pi}} \sqrt{x}, \\ D^{1/2}[x^2] &= \frac{\Gamma(3)}{\Gamma(\frac{5}{2})} x^{3/2} = \frac{2}{\frac{3}{4}\sqrt{\pi}} x^{3/2} = \frac{8}{3\sqrt{\pi}} x^{3/2}. \end{aligned}$$

Check consistency: $D^{1/2} D^{1/2}[x] = D^1[x] = 1$. Verify: $D^{1/2}[\frac{2}{\sqrt{\pi}} \sqrt{x}] = \frac{2}{\sqrt{\pi}} \cdot \frac{\Gamma(\frac{3}{2})}{\Gamma(1)} x^0 = \frac{2}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2} = 1$. ✓

Example 6.2 (Riesz OFI on a Gaussian). Let $f(x) = e^{-\pi x^2}$ (the self-dual Gaussian under Fourier transform: $\hat{f}(\xi) = e^{-\pi \xi^2}$).

The Riesz potential in $\mathbb{R}^1$ at order α has Fourier multiplier $|\xi|^{-\alpha}$, so:

$$\widehat{I^\alpha f}(\xi) = |\xi|^{-\alpha} e^{-\pi \xi^2}.$$

At $\alpha = \frac{1}{2}$: the OFI smooths the Gaussian by a factor $|\xi|^{-1/2}$ in frequency space, suppressing high frequencies (short scales) while preserving low frequencies (large scales) – the fractional smoothing characteristic of $\alpha = \frac{1}{2}$.

The half-derivative $(-\Delta)^{1/4} f$ has multiplier $|\xi|^{1/2}$:

$$\widehat{(-\Delta)^{1/4} f}(\xi) = |\xi|^{1/2} e^{-\pi \xi^2}.$$

This *enhances* high frequencies. The L^2 norm:

$$\|(-\Delta)^{1/4} f\|_{L^2}^2 = \int_{\mathbb{R}} |\xi| e^{-2\pi\xi^2} d\xi = \frac{1}{2\pi} < \infty.$$

The Gaussian is in $\dot{H}^{1/2}(\mathbb{R})$ with $\|f\|_{\dot{H}^{1/2}} = (2\pi)^{-1/2}$.

Physical meaning: In Navier-Stokes, the vorticity ω plays the role of f . Bounded $\|(-\Delta)^{1/4}\omega\|_{L^2}$ means $\omega \in \dot{H}^{1/2}$, which (via Biot–Savart + Prodi–Serrin) prevents blow-up. The Gaussian example shows concretely what “half-derivative is finite” looks like.

7 The Extended Trigonometric Family

7.1 The six primary functions and their OFI orbit

The six standard trigonometric functions arise from two independent functions $(\sin \theta, \cos \theta)$ and their ratios. Under fractional differentiation of order α :

$$D^\alpha[\sin \theta] = \sin\left(\theta + \frac{\alpha\pi}{2}\right).$$

This single formula generates the entire primary family:

α	$D^\alpha[\sin \theta]$	Name
0	$\sin \theta$	sine
1	$\cos \theta$	cosine
2	$-\sin \theta$	minus sine
3	$-\cos \theta$	minus cosine
4	$\sin \theta$	(period 4)
-1	$-\cos \theta$	(integral of sine)

The fractional family for $\alpha \in [0, 1)$ interpolates continuously between sin and cos: every fractional derivative of sin is a phase-shifted sine wave, still periodic, still bounded. This is the OFI orbit of sin.

7.2 Key identities at $\theta = 0$ (small-angle regime)

At $\theta \rightarrow 0$, to first order:

$$\sin \theta \approx \theta, \quad \tan \theta \approx \theta, \quad \cos \theta \approx 1, \quad \sec \theta \approx 1.$$

Thus $\sin \theta \approx \tan \theta \approx \theta$ and $\cos \theta \approx 1$. This is the *centering principle* at the origin: near 0, all odd trig functions collapse to the identity map, and all even trig functions collapse to 1.

The fundamental product identity:

$$\tan \theta \cdot \cot \theta = 1 \quad \forall \theta.$$

$\tan$ covers $(-\frac{\pi}{2}, \frac{\pi}{2}) \rightarrow \mathbb{R}$ (full real line on one period). $\cot$ covers $(0, \pi) \rightarrow \mathbb{R}$. Together, they tile $\mathbb{R}$ with period π , and their product is always 1 — a discrete analog of the Riesz semigroup $I^\alpha I^{-\alpha} = I^0 = \text{Id}$.

7.3 The full extended family

The pre-modern (navigational) trigonometric family extends the six primary functions to cover every ratio and complement:

Name	Symbol	Definition
<i>Primary (6 functions)</i>		
sine	$\sin \theta$	opposite/hypotenuse
cosine	$\cos \theta$	adjacent/hypotenuse
tangent	$\tan \theta$	$\sin \theta / \cos \theta$
cotangent	$\cot \theta$	$\cos \theta / \sin \theta$
secant	$\sec \theta$	$1 / \cos \theta$
cosecant	$\csc \theta$	$1 / \sin \theta$
<i>Versed (4 functions)</i>		
versine	$\text{vers } \theta$	$1 - \cos \theta$
coversine	$\text{covers } \theta$	$1 - \sin \theta$
vercosine	$\text{verc } \theta$	$1 + \cos \theta$
covercosine	$\text{coverc } \theta$	$1 + \sin \theta$
<i>Half-versed (4 functions)</i>		
haversine	$\text{hav } \theta$	$\frac{1}{2}(1 - \cos \theta) = \sin^2(\theta/2)$
hacoversine	$\text{hacov } \theta$	$\frac{1}{2}(1 - \sin \theta)$
havercosine		$\frac{1}{2}(1 + \cos \theta) = \cos^2(\theta/2)$
hacovercosine		$\frac{1}{2}(1 + \sin \theta)$
<i>Exterior (2 functions)</i>		
exsecant	$\text{exsec } \theta$	$\sec \theta - 1$
excosecant	$\text{excsc } \theta$	$\csc \theta - 1$
<i>Chord functions (historical)</i>		
chord	$\text{crd } \theta$	$2 \sin(\theta/2)$

7.4 OFI unification of the trig family

All of these functions are generated by applying operations (complement, reciprocal, shift, half-argument) to $\sin$ and $\cos$. Under OFI:

- $\sin$ and $\cos$ are related by $D^1[\sin] = \cos$ (one full derivative).
- $D^{1/2}[\sin \theta] = \sin(\theta + \pi/4) = \frac{\sqrt{2}}{2}(\sin \theta + \cos \theta)$ — the half-derivative of sine is the *diagonal* combination.
- $\text{hav } \theta = \sin^2(\theta/2)$ is the half-argument squared — the L^2 norm of the half-frequency component, directly analogous to the OFI energy $\|(-\Delta)^{1/4} f\|_{L^2}^2$.
- $\text{vers } \theta = 1 - \cos \theta = 2 \sin^2(\theta/2)$ is twice the haversine — the “defect from 1”, parallel to the OFI defect $R_{\text{loc}}^{1/2}$ measuring how far the solution is from smooth.

The centering principle (Section 5) appears throughout: haversine uses the half-angle $\theta/2$ (midpoint of $[0, \theta]$), exsecant measures the excess $\sec \theta - 1$ (deviation from the identity at $\theta = 0$). Every extended trig function is a measurement of deviation from the centered,

symmetric baseline — exactly as the Riesz kernel measures deviation from the distributional identity $\delta(x - y)$.

8 OFI on the Standard Model Lagrangian

8.1 The Standard Model Lagrangian

The Standard Model (SM) Lagrangian density is:

$$\mathcal{L}_{\text{SM}} = \underbrace{-\frac{1}{4}F_{\mu\nu}^a F^{a\mu\nu}}_{\text{Yang-Mills (gauge)}} + \underbrace{\bar{\psi}(i\gamma^\mu D_\mu - m)\psi}_{\text{Dirac (matter)}} + \underbrace{|D_\mu\phi|^2 - V(\phi)}_{\text{Higgs}} + \underbrace{\mathcal{L}_{\text{Yukawa}}}_{\text{matter-Higgs}}, \quad (1)$$

where $F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + gf^{abc}A_\mu^b A_\nu^c$ is the gauge field strength, $D_\mu = \partial_\mu + igA_\mu^a T^a$ is the covariant derivative, ψ is a spinor field (quarks/leptons), ϕ is the Higgs field.

8.2 Running OFI on the Lagrangian

Define the localized OFI defect for a field F at scale a on ball B :

$$R^{1/2}(F, a, B) := \|(-\Delta)^{1/4}(K_a * F)\|_{L^2(B)},$$

where K_a is the boxcar as an **approximate identity only** (NOT the fractional part), and $(-\Delta)^{1/4}$ is the genuine Riesz derivative of order $1/2$ with Fourier multiplier $|\xi|^{1/2}$.

Sector by sector:

Sector	Field F	OFI defect
Yang-Mills	$F_{\mu\nu}^a$ (gluon/gauge curvature)	$R_{\text{YM}}^{1/2}(A, a, B)$
Dirac	ψ (quark/lepton)	$R_{\text{Dirac}}^{1/2}(\psi, a, B)$
Higgs	ϕ	$R_{\text{Higgs}}^{1/2}(\phi, a, B)$
Navier-Stokes (fluid analogy)	$\omega = \nabla \times u$ (vorticity)	$R_{\text{loc}}^{1/2}(u, a, B)$

8.3 Yang-Mills and the mass gap

The Yang-Mills sector is:

$$\mathcal{L}_{\text{YM}} = -\frac{1}{4}F_{\mu\nu}^a F^{a\mu\nu}.$$

The canonical OFI defect is $R_{\text{YM}}^{1/2}(A, a, B) = \|(-\Delta)^{1/4}(K_a * F_{\mu\nu})\|_{L^2(B)}$.

Conjecture 8.1 (OFI mass-gap, Riesz formulation). There exists $\delta > 0$ and scale $a_0 > 0$ such that for every Yang-Mills connection A :

$$\inf_{A \in \text{YM}} \|R_{\text{YM}}^{1/2}(A, a_0, \mathbb{R}^4)\|_{L^2} \geq \delta(a_0, g) > 0.$$

This says the OFI defect of the gauge curvature cannot be driven to zero (there is a floor). Physically: the Yang-Mills field cannot be made arbitrarily smooth at any scale — it always has a minimum “fractional roughness” set by δ . This floor is the mass gap.

8.4 Why boxcar fails here, and why OFI does not

A boxcar (moving average) of $F_{\mu\nu}$ can be made arbitrarily small by choosing a large: $K_a * F_{\mu\nu} \rightarrow \langle F_{\mu\nu} \rangle \rightarrow 0$ for pure-gauge configurations. So a boxcar-based “defect” has no floor.

The Riesz derivative $(-\Delta)^{1/4}$, by contrast, *amplifies* high-frequency content (Fourier multiplier $|\xi|^{1/2}$ grows with $|\xi|$). For a non-trivial Yang–Mills connection the high-frequency curvature is topologically protected — it cannot vanish — so the OFI defect has a genuine lower bound. This is why the semigroup property is essential: $I^\alpha I^\beta = I^{\alpha+\beta}$ ensures that consecutive OFI measurements at different scales compose correctly, preserving the topological floor.

8.5 Mass vs. gravity: the OFI perspective

Property	Mass (inertial/gravitational)	Gauge mass gap
Origin	Higgs mechanism (ϕ VEV)	Dimensional transmutation
Formula	$m_f = y_f v / \sqrt{2}$ (Yukawa)	$m \sim \Lambda_{\text{QCD}} e^{-8\pi^2/(g^2 N)}$
OFI defect	$R_{\text{Higgs}}^{1/2} > 0$ (SSB)	$R_{\text{YM}}^{1/2} \geq \delta > 0$ (floor)
Scale	Electroweak ($v = 246$ GeV)	QCD ($\Lambda_{\text{QCD}} \approx 200$ MeV)
Gravity	Curved spacetime (GR, $g_{\mu\nu}$)	Gauge curvature $F_{\mu\nu}$

Gravity couples to the stress-energy tensor $T_{\mu\nu}$, which involves all fields. In OFI language, gravity is sensitive to the *total* OFI defect across all sectors:

$$R_{\text{gravity}}^{1/2} \sim R_{\text{YM}}^{1/2} + R_{\text{Higgs}}^{1/2} + R_{\text{Dirac}}^{1/2} + \dots$$

The mass gap (minimum OFI floor in the YM sector) and gravitational coupling are independent: mass is a Higgs VEV, gravity couples to all of $T_{\mu\nu}$.

9 Relativistic OFI: Spin-2, Spacetime Geometry, and Vacuum Energy

9.1 Spin and the extended trig family

In quantum field theory, a particle of spin s transforms under a rotation by angle ϕ about its propagation axis as

$$\Phi^{(s)} \rightarrow e^{is\phi} \Phi^{(s)}.$$

The two physical (transverse) polarization modes of a spin- s boson have helicities $m = \pm s$. Writing them as real combinations:

$$h_+ \rightarrow \cos(s\phi) h_+ - \sin(s\phi) h_\times, \quad h_\times \rightarrow \sin(s\phi) h_+ + \cos(s\phi) h_\times.$$

The full trig family encodes every spin:

Spin s	Particle	Rotation uses	OFI order	Trig connection
0	Higgs scalar	1 (trivial)	$\alpha = 0$	identity
$\frac{1}{2}$	Quark/lepton	$\cos(\phi/2), \sin(\phi/2)$	$\alpha = \frac{1}{2}$	half-angle: $\text{hav}(\phi) = \sin^2(\phi/2)$
1	Photon, $W^\pm$, Z , gluon	$\cos \phi, \sin \phi$	$\alpha = 1$	standard trig
2	Graviton	$\cos(2\phi), \sin(2\phi)$	$\alpha = 2$	double-angle / haversine

Spin-2 in the extended trig family. The graviton (spin-2) rotation uses double-angle formulas, which decompose directly into the extended trig family:

$$\begin{aligned}
\cos(2\phi) &= 1 - 2\sin^2 \phi = 1 - 2 \text{hav}(2\phi) = 1 - \text{vers}(2\phi), \\
\sin(2\phi) &= 2 \sin \phi \cos \phi, \\
1 - \cos(2\phi) &= \text{vers}(2\phi) = 2 \text{hav}(2\phi) = 2 \sin^2 \phi.
\end{aligned}$$

The haversine $\text{hav}(2\phi)$ and versine $\text{vers}(2\phi)$ are the natural coordinates for spin-2 rotations. The graviton polarization at angle $\phi = 45^\circ$ gives $\text{hav}(90^\circ) = \frac{1}{2}$, $\text{vers}(90^\circ) = 1$ — the $\times$ -mode is fully mixed into the $+$ -mode.

The OFI fractional derivative order $\alpha = s$ encodes spin: $D^s[\sin \theta] = \sin(\theta + s\pi/2)$ rotates the phase by $s \cdot 90^\circ$. Spin-2 shifts phase by 180° : $D^2[\sin \theta] = -\sin \theta$, so two graviton polarizations are π -apart, consistent with 180° periodicity of the polarization tensor $h_{\mu\nu}$.

9.2 de Sitter, anti-de Sitter, and Minkowski spacetimes

Einstein's field equations with cosmological constant Λ :

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}.$$

The three maximally symmetric vacuum solutions ($T_{\mu\nu} = 0$) are:

Spacetime	Λ	Curvature	Isometry	OFI defect	Trig geometry
Minkowski	0	0 (flat)	Poincaré ISO(1,3)	$R^{1/2} = 0$	Euclidean $\sin, \cos$
de Sitter (dS)	> 0	$+\frac{4\Lambda}{3}$	SO(1,4)	$R^{1/2} \sim \sqrt{\Lambda/3}$	Hyperbolic $\sinh, \cosh$
Anti-de Sitter (AdS)	< 0	$-\frac{4 \Lambda }{3}$	SO(2,3)	$R^{1/2} \sim \sqrt{ \Lambda /3}$	Oscillatory (AdS box)

Minkowski is the zero of the OFI scale. The Riesz kernel $|x - y|^{\alpha-n} \rightarrow \delta(x - y)$ as curvature $\rightarrow 0$; the OFI defect $R_{\text{grav}}^{1/2} = 0$ reflects the absence of curvature fluctuations. All trig functions evaluate at their origin-values: $\sin(0) = 0$, $\cos(0) = 1$.

de Sitter ($\Lambda > 0$) is our universe's late-time attractor (dark energy dominates). The expansion scale factor is $a(t) = e^{Ht}$ where $H = \sqrt{\Lambda/3}$ is the Hubble rate. In OFI:

$$R_{\text{ds}}^{1/2}(a, B) = \|(-\Delta)^{1/4}(K_a * R_{\mu\nu\rho\sigma})\|_{L^2(B)} \sim \sqrt{\Lambda/3},$$

a uniform, positive defect throughout spacetime — consistent with the uniform positive curvature of dS. The de Sitter metric uses hyperbolic trig: $ds^2 = -dt^2 + \cosh^2(Ht)(d\Omega_3^2)$

(global slicing), connecting the extended hyperbolic family ($\sinh$, $\cosh$, $\tanh$, ...) to the positively-curved OFI regime.

Anti-de Sitter ($\Lambda < 0$) is the arena of the AdS/CFT correspondence (Maldacena 1997): a $d + 1$ -dimensional gravitational theory in AdS is equivalent to a d -dimensional conformal field theory on its boundary. In OFI:

$$R_{\text{AdS}}^{1/2}(a, B) \sim \sqrt{|\Lambda|/3} > 0,$$

but with *negative* sign of Λ creating an effective potential well — AdS is like a box. Fields in AdS oscillate (standard trig) rather than exponentially grow (hyperbolic trig), placing AdS in the $\sin/\cos$ branch of the extended family.

Summary (trig branch by sign of Λ):

$\Lambda < 0$: oscillatory (standard trig) $\Lambda = 0$: linear (identity) $\Lambda > 0$: exponential (hyperbolic trig).

This is the OFI analog of the centering principle: Minkowski is the midpoint ($\Lambda = 0$), de Sitter and anti-de Sitter are the $\pm$ branches.

9.3 Black holes vs. dark matter: OFI sectors and gravitational collapse

Black holes form when a matter distribution collapses past its Schwarzschild radius $r_s = 2GM/c^2$. Gravitational collapse is possible because:

- Baryonic matter radiates energy (EM interaction: $R_{\text{EM}}^{1/2} > 0$), shedding angular momentum and allowing collapse.
- Radiation pressure ($R_{\text{Higgs}}^{1/2}$, $R_{\text{Dirac}}^{1/2}$) eventually fails when the star runs out of nuclear fuel.
- The *total* OFI defect $R_{\text{total}}^{1/2} = R_{\text{YM}}^{1/2} + R_{\text{EM}}^{1/2} + R_{\text{Higgs}}^{1/2} + R_{\text{grav}}^{1/2}$ allows energy dissipation through multiple channels.

Dark matter cannot undergo the same collapse because its OFI defect has only a gravitational component:

$$R_{\text{DM}}^{1/2} = R_{\text{grav}}^{1/2} \quad \text{only} \quad (\text{no EM, no QCD, possibly weak}).$$

Without electromagnetic interaction, dark matter cannot radiate — it cannot shed energy during collapse. It forms extended, diffuse halos (not compact objects) because:

1. No cooling: $R_{\text{EM}}^{1/2} = 0$ means no photon emission to carry away kinetic energy.
2. No Pauli exclusion in the same form: without QCD confinement, no degeneracy pressure from color-singlet packing.
3. The Jeans instability applies but on cosmological timescales ($\sim 10^{10}$ years), not stellar timescales ($\sim 10^7$ years).

Gravitational interaction between black holes and dark matter. Both couple to the metric $g_{\mu\nu}$ via the stress-energy tensor $T_{\mu\nu}$. The spin-2 graviton mediates this interaction for all forms of energy-momentum. In OFI:

$$\langle R_{\text{grav}}^{1/2}(A_{\text{BH}}), R_{\text{grav}}^{1/2}(A_{\text{DM}}) \rangle \neq 0.$$

Dark matter halos provide the gravitational potential ($R_{\text{DM}}^{1/2} > 0$) that black holes sit in; black hole accretion (spin-2 mediated) can capture dark matter that passes within the capture radius $r_c = r_s \sqrt{1 + v_\infty^2/c^2}$. Conversely, black holes cannot “eat” dark matter efficiently precisely because $R_{\text{EM,DM}}^{1/2} = 0$: there is no friction or dissipation to slow dark matter into a capture orbit. The black hole sees the dark matter only through the gravitational channel.

Property	Black hole	Dark matter halo
OFI sectors active	All: grav + EM + QCD + Higgs	Gravity only
Collapse mechanism	Radiative cooling (EM) + gravity	Gravity only, no cooling
Compact object?	Yes ($r \lesssim r_s$)	No (diffuse, $r \sim \text{kpc}$)
Interact with each other?	Yes (spin-2 graviton)	Yes (spin-2 graviton)
Radiate?	Yes (Hawking, EM synchrotron)	No (no EM coupling)
OFI defect floor	$R^{1/2} \geq c_{\text{BH}} M^{-1/2}$	$R_{\text{DM}}^{1/2} = R_{\text{grav}}^{1/2}$ only

9.4 The $-\frac{1}{12}$, vacuum energy, tachyon condensation, and Zwiebach

9.4.1 String theory and the critical dimension

In light-cone quantization of the bosonic string in D spacetime dimensions, there are $D - 2$ transverse oscillators. Each bosonic mode n contributes zero-point energy $-n/2$ (from normal ordering : $aa^\dagger := a^\dagger a$). The total normal-ordering constant is:

$$a = -\frac{D-2}{2} \sum_{n=1}^{\infty} n = -\frac{D-2}{2} \zeta(-1) = -\frac{D-2}{2} \cdot \left(-\frac{1}{12}\right) = \frac{D-2}{24}.$$

The open-string mass formula is $\alpha' m^2 = N - a$, where N is the oscillator level. For a massless spin-1 state at level $N = 1$:

$$a = 1 \implies \frac{D-2}{24} = 1 \implies \boxed{D = 26}.$$

The $-\frac{1}{12}$ from the regularized sum $1 + 2 + 3 + \dots = \zeta(-1) = -\frac{1}{12}$ directly forces the critical dimension of bosonic string theory to be $D = 26$. Remove the $-\frac{1}{12}$ (use a boxcar cutoff instead), and the critical dimension is lost. This is the most physically consequential appearance of $\zeta(-1) = -\frac{1}{12}$ in all of physics.

For the superstring: each fermionic oscillator contributes $+n/2$ to the zero-point energy (opposite sign to bosons via the spin-statistics theorem). After cancellations (worldsheet supersymmetry), the critical dimension reduces to $D = 10$.

9.4.2 Tachyon condensation and Zwiebach

The open bosonic string at level $N = 0$ has:

$$\alpha' m^2 = 0 - 1 = -1 \implies m^2 = -\frac{1}{\alpha'} < 0.$$

A field with $m^2 < 0$ is a **tachyon**: it sits at the top of a potential hill, not a minimum. The perturbative bosonic string vacuum is unstable.

Sen’s conjecture (1999): the open-string tachyon can roll to a new stable minimum (“tachyon condensate”) where the D-brane has annihilated. The energy difference between the two vacua exactly equals the D-brane tension:

$$V(T_0) - V(0) = -T_{Dp}.$$

Zwiebach’s contribution: Barton Zwiebach (MIT) formulated *open string field theory* (OSFT), where the tachyon becomes a proper field Φ with an action functional. Using *level truncation* — keeping only finitely many string modes — Zwiebach and collaborators verified Sen’s conjecture to high precision:

Level truncation	$V(T_0)/T_{Dp}$ computed	Error
Level 0	−0.684	32%
Level 2	−0.950	5%
Level 4	−0.986	1.4%
Level 10	−0.9994	0.06%

The convergence to -1 (exact Sen conjecture) is a non-trivial confirmation that the $-\frac{1}{12}$ normal-ordering constant is correct.

9.4.3 OFI interpretation of the vacuum and tachyon

In the OFI framework, the pair $(-1, \frac{1}{12})$ encodes the vacuum structure:

State	m^2	OFI defect interpretation
Tachyon ($N = 0$)	$-1/\alpha' < 0$	$R^{1/2} < 0$ (imaginary): below the floor, unstable
Massless ($N = 1$)	0	$R^{1/2} = 0$: on the floor, conformally invariant
Massive ($N \geq 2$)	> 0	$R^{1/2} > 0$: above the floor, stable
Tachyon condensate	$V = -T_{Dp}$	$R^{1/2} = 0$: D-brane annihilated, new stable vacuum

The tachyon $m^2 = -1/\alpha'$ has the same -1 as in the pair $(-1, \frac{1}{12})$:

- The -1 is the *pole order* of the divergence in the vacuum energy (from $\zeta(-1)$).
- The $\frac{1}{12}$ is the *finite regularized value*.
- The tachyon mass $m^2 = -1/\alpha'$ is the *physical consequence* of the $a = 1$ normal-ordering shift derived from $\zeta(-1) = -\frac{1}{12}$.

Tachyon condensation as OFI defect cancellation. The tachyon rolls from $R^{1/2} < 0$ (unstable, below-floor) to $R^{1/2} = 0$ (new vacuum, floor reached). This is precisely the OFI analog of nucleon–anti-nucleon annihilation (Section ??): two OFI defects with opposite signs cancel, releasing energy equal to the “brane tension” (the absolute value of the negative defect).

The Casimir vacuum energy. Between two conducting plates separated by distance L , the quantum vacuum (zero-point oscillations) generates an attractive force. The energy:

$$E_{\text{Casimir}} = -\frac{\pi^2}{720} \frac{\hbar c}{L^3},$$

where the $\pi^2/720 = \pi^2/(720)$ comes from $\zeta(-3) = -\frac{1}{120}$ and $\zeta(-1) = -\frac{1}{12}$ via the same normal-ordering regularization. In OFI language, the Casimir energy is the integrated vacuum OFI defect:

$$E_{\text{Casimir}} = -c_{3,1/2} \int_0^L R_{\text{vac}}^{1/2}(a, x) dx,$$

where the floor set by $-\frac{1}{12}$ per oscillator mode integrates to the observed attractive force. The pair $(-1, \frac{1}{12})$ is not abstract — it is measurable to 1 part in 10^3 with current experiments.

10 Hybridization and Atomic Orbitals via OFI

10.1 Atomic orbitals and angular momentum

A chemist or physicist will recognize atomic orbitals from quantum mechanics:

$$\psi_{n,\ell,m}(r, \theta, \varphi) = R_{n\ell}(r) Y_{\ell}^m(\theta, \varphi),$$

where $R_{n\ell}$ is the radial function and Y_{ℓ}^m is the spherical harmonic. The quantum number ℓ labels the orbital type:

ℓ	Orbital	Shape	Angular momentum L^2 eigenvalue
0	s	spherically symmetric	$\ell(\ell+1)\hbar^2 = 0$
1	p	dumbbell (directional)	$\ell(\ell+1)\hbar^2 = 2\hbar^2$
2	d	cloverleaf	$\ell(\ell+1)\hbar^2 = 6\hbar^2$
3	f	complex polylobes	$\ell(\ell+1)\hbar^2 = 12\hbar^2$

10.2 Hybridization as fractional mixing

Hybridization mixes orbitals of different ℓ . In carbon ($Z = 6$):

- sp^3 : one s + three $p \rightarrow$ four equivalent orbitals, 109.5°
- sp^2 : one s + two $p \rightarrow$ three, 120° , one unhybridized p
- sp : one s + one $p \rightarrow$ two, 180° , two unhybridized p

Definition 10.1 (Hybridization OFI parameter). Let $\alpha \in [0, 1]$ be the fractional OFI order interpolating between s ($\alpha = 0$) and p ($\alpha = 1$) character. The hybrid orbital at order α is:

$$\psi_{\alpha} := I^{\alpha} \psi_s = c_{3,\alpha} \int_{\mathbb{R}^3} |x - y|^{\alpha-3} \psi_s(y) dy,$$

where $\psi_s = R_{10}(r)Y_0^0$ is the $1s$ orbital and I^{α} is the 3D Riesz potential of order α .

The Riesz kernel $|x - y|^{\alpha-3}$ interpolates the spatial non-locality:

- At $\alpha = 0$: $I^0 = \text{Id}$ — pure s character (spherical, local).
- At $\alpha = 1$: $I^1 = \text{Coulomb potential kernel}$ ($|x - y|^{-2}$ in 3D) — directional, p -like.
- At $\alpha \in (0, 1)$: intermediate hybridization.

The hybridization ratio in sp^k is $\alpha = k/(k + 1)$:

$$\text{sp} : \alpha = \frac{1}{2}, \quad \text{sp}^2 : \alpha = \frac{2}{3}, \quad \text{sp}^3 : \alpha = \frac{3}{4}.$$

Theorem 10.2 (Bond angle from OFI parameter). *The bond angle ϕ_α of the sp^α hybrid satisfies:*

$$\cos \phi_\alpha = -\frac{\alpha}{1 - \alpha} \implies \phi_\alpha = \arccos\left(-\frac{\alpha}{1 - \alpha}\right).$$

Particular values:

$$\begin{aligned} \alpha = \frac{1}{2} &\Rightarrow \cos \phi = -1 \Rightarrow \phi = 180^\circ && (\text{sp}, \text{acetylene}) \\ \alpha = \frac{2}{3} &\Rightarrow \cos \phi = -\frac{1}{2} \Rightarrow \phi = 120^\circ && (\text{sp}^2, \text{ethylene}) \\ \alpha = \frac{3}{4} &\Rightarrow \cos \phi = -\frac{1}{3} \Rightarrow \phi = 109.47^\circ && (\text{sp}^3, \text{methane}) \end{aligned}$$

This is exact: the standard hybridization bond angles are reproduced by the OFI parameter $\alpha = k/(k + 1)$ with no fitting.

10.3 Ionic vs. covalent bonds: classical vs. fractional entanglement

Property	Ionic bond	Covalent bond
Mechanism	Full electron transfer	Electron sharing
Quantum state	Product state: $ A^+\rangle B^-\rangle$	Entangled: $\frac{1}{\sqrt{2}}(A \uparrow B \downarrow\rangle - A \downarrow B \uparrow\rangle)$
OFI parameter	$\alpha \rightarrow 0$ (local, no fractional mixing)	$\alpha \rightarrow 1$ (non-local, full Riesz)
OFI defect R^α	$\rightarrow 0$ as $\alpha \rightarrow 0$	$R^{1/2} > 0$ (non-trivial)
Analogy	Boxcar (zero semigroup mixing)	Riesz (genuine semigroup)

Covalent bond = quantum entanglement = non-zero OFI defect. When two atoms share an electron, the electron wavefunction cannot be written as a product of individual atomic states — it is entangled. The OFI defect $R^{1/2} > 0$ measures this non-factorizability: a non-zero half-derivative means the wavefunction has genuine fractional correlations between the two nuclei.

An ionic bond ($\alpha \approx 0$) is a classical correlation — you know the electron is on B, not A. No entanglement, no OFI defect.

Bond order and OFI order:

$$\begin{aligned} \text{Single bond } (\sigma): & \quad \text{OFI order } \alpha = \frac{1}{2} \\ \text{Double bond } (\sigma + \pi): & \quad \text{OFI order } \alpha = \frac{2}{3} \\ \text{Triple bond } (\sigma + 2\pi): & \quad \text{OFI order } \alpha = \frac{3}{4} \end{aligned}$$

The higher the bond order, the more electrons are shared, the larger α , the stronger the fractional non-locality of the OFI kernel.

11 Nucleons, Anti-Nucleons, and Gluons

11.1 QCD and the strong force

Quantum chromodynamics (QCD) is the $SU(3)$ gauge theory of the strong force. The Lagrangian:

$$\mathcal{L}_{\text{QCD}} = -\frac{1}{4}G_{\mu\nu}^a G^{a\mu\nu} + \sum_f \bar{q}_f (i\gamma^\mu D_\mu - m_f) q_f,$$

where $G_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + g_s f^{abc} A_\mu^b A_\nu^c$ is the gluon field strength (8 gluons for $SU(3)$), q_f are quark fields with flavors $f \in \{u, d, s, c, b, t\}$, and $D_\mu = \partial_\mu - ig_s A_\mu^a T^a$ is the covariant derivative (T^a are $SU(3)$ generators).

11.2 Nucleon and anti-nucleon structure

Particle	Quark content	Charge	Baryon number
Proton	uud	+1	+1
Neutron	udd	0	+1
Anti-proton	$\bar{u}\bar{u}\bar{d}$	-1	-1
Anti-neutron	$\bar{u}\bar{d}\bar{d}$	0	-1

Gluons carry *color charge* (red, green, blue and their anti-colors). Quarks are confined: free quarks are never observed. The energy cost of separating two quarks grows linearly with separation (“string tension”), a consequence of the non-Abelian self-interaction of gluons ($f^{abc} \neq 0$ in $SU(3)$, unlike QED where $f^{abc} = 0$ for $U(1)$).

11.3 OFI treatment of QCD

The canonical QCD OFI defect at scale a on ball B :

$$R_{\text{QCD}}^{1/2}(A, a, B) := \|(-\Delta)^{1/4}(K_a * G_{\mu\nu}^a)\|_{L^2(B)},$$

where the sum is over the 8 gluon components $a = 1, \dots, 8$.

Key features of the QCD OFI defect:

1. **Confinement floor:** For a nucleon (3-quark bound state), the gluon field $G_{\mu\nu}^a$ carries topological information (the baryon number Wess–Zumino term). By the topological OFI floor theorem:

$$R_{\text{QCD}}^{1/2}(A_{\text{nucleon}}, a, B) \geq c(g_s) a^{-1/2} \sqrt{N_c},$$

where $N_c = 3$ is the number of colors. The gluon field can never be made smooth — confinement as an OFI lower bound.

2. **Nucleon–anti-nucleon annihilation:** When a proton (uud) meets an anti-proton ($\bar{u}\bar{u}\bar{d}$), quark–anti-quark pairs annihilate. In OFI language, $R^{1/2}(A_p + A_{\bar{p}}, a, B)$: the two OFI defects *cancel* (color-charge cancellation), driving the defect to zero. The energy released ($\sim 2m_p c^2 = 1876$ MeV) corresponds to the jump from $R^{1/2} > 0$ (confined quarks) to $R^{1/2} \approx 0$ (annihilated, free energy emitted as pions/photons).

3. **Gluon self-interaction and the semigroup:** The non-Abelian $f^{abc}A_\mu^b A_\nu^c$ term means gluons interact with each other — unlike photons. In OFI terms, the gluon OFI satisfies $I^\alpha I^\beta = I^{\alpha+\beta}$ (semigroup), but the interaction vertex couples different OFI scales. This multi-scale coupling is the OFI signature of asymptotic freedom: at short scales ($a \rightarrow 0$, high $|\xi|$), the coupling $g_s(|\xi|) \rightarrow 0$; at long scales, $g_s \rightarrow \infty$ (confinement).
4. **OFI and hybridization of color:** Just as sp^3 hybridization mixes 4 atomic orbitals, $SU(3)$ color combines 3 color states into a color-singlet baryon. The OFI parameter $\alpha = 2/3$ (as in sp^2) matches the 3-quark color averaging: each quark carries $\alpha = 1/3$ of the total OFI interaction, and $3 \times \frac{1}{3} = 1$ gives the fully symmetric color-singlet state.

11.4 Connecting hybridization to nucleon structure

The analogy between atomic hybridization and quark confinement is not merely metaphorical:

Concept	Chemistry	Nuclear physics
Gauge group	$U(1)$ (EM)	$SU(3)$ (color)
“Orbital”	Atomic orbital $\psi_{n\ell m}$	Color wavefunction $ r\rangle, g\rangle, b\rangle$
“Hybridization”	sp^3 , bond angle 109.5°	Color singlet $\epsilon^{ijk} rgb\rangle/\sqrt{6}$
“Bond”	Covalent (shared e^-)	Gluon flux tube (shared color)
“Ionic”	Full e^- transfer	Free quark (impossible in QCD)
OFI order	$\alpha = k/(k+1)$	$\alpha = (N_c - 1)/N_c = 2/3$
OFI defect floor	$R_{\text{bond}}^{1/2} > 0$ (covalent)	$R_{\text{QCD}}^{1/2} \geq c a^{-1/2}$ (confinement)

The color-singlet condition (quarks must form a color-neutral hadron) is the nuclear analog of the hybridization condition (orbitals must sum to a net-neutral electron configuration). In both cases, the OFI framework provides the unifying language: a fractional non-local kernel measuring how “mixed” the constituent states are.

12 Summary and Honest Status

12.1 What OFI gives you, precisely

1. **The Gamma function extension:** $(-1)! = \Gamma(0) = \infty$ is the mechanism that forces $I^0 = \text{Id}$. Not a bug.
2. **The ordering problem, resolved:** The symmetric Riesz kernel has no preferred endpoint, so $D^\alpha I^\alpha = I^\alpha D^\alpha = \text{Id}$.
3. **The semigroup law:** $I^\alpha I^\beta = I^{\alpha+\beta}$ (exact, proved by Fourier multipliers). Boxcar fails this.
4. **Regularization pairs:** Divergent quantities carry two numbers: (pole structure, finite part) $= (-1, \frac{1}{12})$, not just $-\frac{1}{12}$.

5. **Fractional trig:** $D^\alpha[\sin \theta] = \sin(\theta + \alpha\pi/2)$ generates the entire extended trig family as an OFI orbit.
6. **NS blow-up criterion:** bounded $R_{\text{loc}}^{1/2} = \|(-\Delta)^{1/4}(K_a * \omega)\|_{L^2} \rightarrow$ no blow-up (via $^{1/2}$ vorticity $\rightarrow$ $^{3/2}$ velocity $\rightarrow$ Prodi–Serrin).
7. **YM mass gap:** the OFI defect $R_{\text{YM}}^{1/2}$ has a topological lower floor for non-trivial gauge connections.
8. **Hybridization:** sp^k bond angle given exactly by $\alpha = k/(k+1)$, $\cos \phi_\alpha = -\alpha/(1-\alpha)$.
9. **Confinement:** color-singlet nucleons have a non-zero OFI defect floor; nucleon–anti-nucleon annihilation is OFI-defect cancellation.

12.2 What remains open

Problem	OFI contribution	Remaining gap
3D NS global regularity	Conditional theorem (if $R^{1/2} \leq M$, no blow-up)	Prove $R^{1/2} \leq M$ for all s
YM mass gap (Clay Prize)	Topological floor, heat-flow regularity	Constructive 4D measure
Color confinement (QCD)	OFI defect floor $\geq c a^{-1/2}$ (classical)	Non-perturbative proof w
Hybridization (exact)	Bond angles from $\alpha = k/(k+1)$ (exact)	First-principles derivation

The single most important sentence: *OFI is not a statistical filter. It is a genuine fractional integral defined by the Riesz kernel $|x-y|^{\alpha-n}$, with Fourier multiplier $|\xi|^{-\alpha}$, satisfying the semigroup law $I^\alpha I^\beta = I^{\alpha+\beta}$. Any approach that replaces this with a boxcar average is doing statistics, not calculus.*